

“Cubix internship task”

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“Cyber security task”

“Maltego transforms”

Run 15 Maltego Transforms on any company of your choice:

Introduction to Maltego

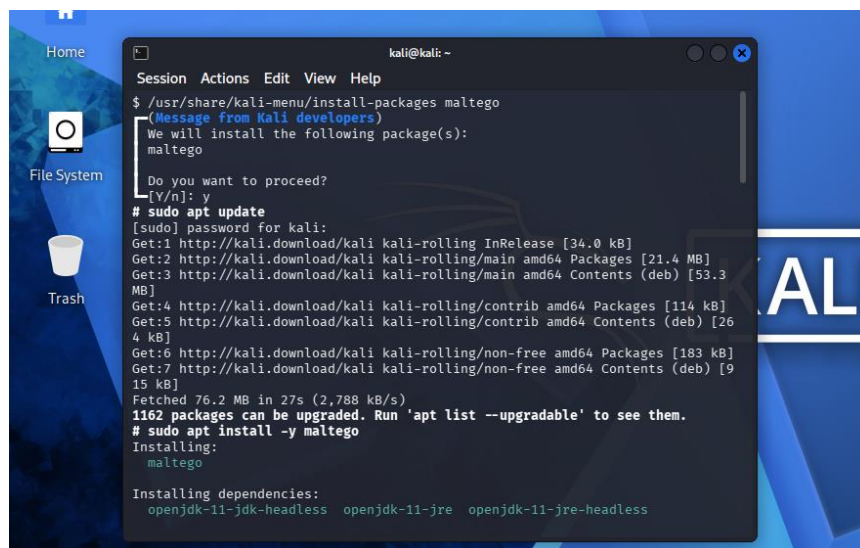
Maltego is an industry-standard Open Source Intelligence (OSINT) and graphical link-analysis software used by cybersecurity professionals to map out digital assets and relationships. By utilizing automated scripts called **transforms**, Maltego queries public databases, DNS records, and threat intelligence APIs to transform a single starting node—such as a domain name—into a comprehensive visual network graph. This capability makes it an indispensable tool for non-intrusive reconnaissance, attack surface management, and footprinting complex enterprise infrastructures during security assessments.

Target Selection: Systems Limited (systemsltd.com)

Selecting Systems Limited as the primary target for this OSINT investigation provides an ideal real-world model for enterprise-level infrastructure mapping. As a premier global technology consulting firm headquartered in Pakistan, Systems Limited operates an expansive digital footprint encompassing multi-region cloud services, custom-configured DNS routing, strict email security mechanisms, and diverse netblock allocations. Reconnaissance on systemsltd.com offers a rich dataset for testing Maltego's transform capabilities, allowing us to map dual-stack IPv4/IPv6 name servers, hybrid cloud mail protection gateways (Microsoft 365), and external netblocks within a real-world corporate attack surface without initiating intrusive network scans.

Solution of the task:

Executed `sudo apt install -y maltego` in terminal. Package manager retrieved dependencies (openjdk-11-jdk-headless, openjdk-11-jre) and configured Maltego 4.12.1.

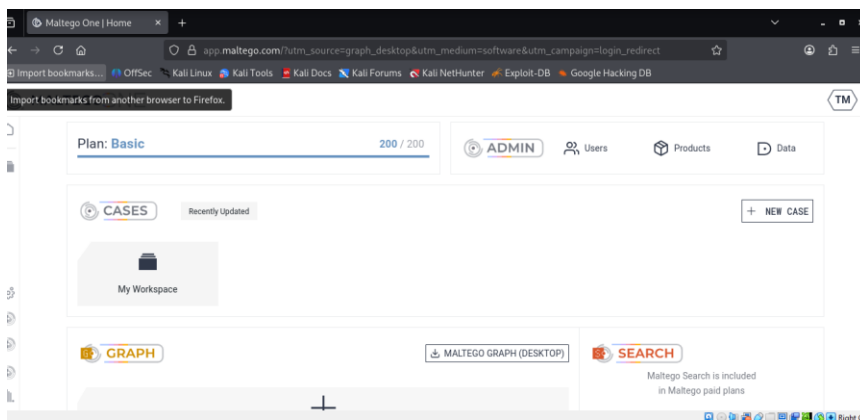


```
kali@kali: ~  
Session Actions Edit View Help  
$ /usr/share/kali-menu/install-packages maltego  
(Message from Kali developers)  
We will install the following package(s):  
maltego  
Do you want to proceed?  
[Y/n]: y  
# sudo apt update  
[sudo] password for kali:  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.3 MB]  
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [264 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]  
Fetched 76.2 MB in 27s (2,788 kB/s)  
1162 packages can be upgraded. Run 'apt list --upgradable' to see them.  
# sudo apt install -y maltego  
Installing:  
maltego  
Installing dependencies:  
openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless
```

Maltego CE splash screen loading core modules in Kali Linux prior to user login



Maltego One web interface indicating account status on the Basic plan with 200/200 credits available before starting reconnaissance.

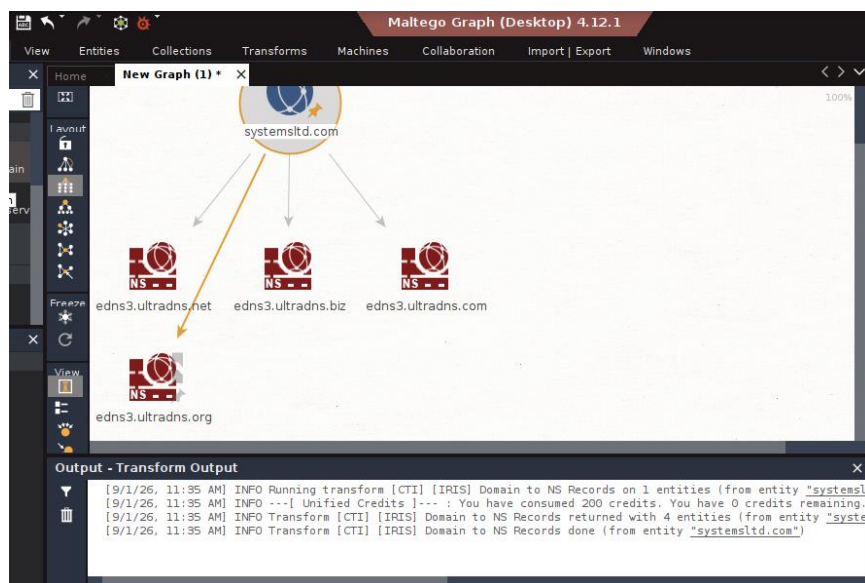


The screenshot displays the EntityRaider application interface. The main window shows a 'New Graph (2)' with a single node labeled 'systemsd.com' and a circular icon. The left sidebar contains a search bar and a list of entity types: Domain, Personal, Registrar, Custom, and ScamAdfuser Tag. The right sidebar shows a 'Detail View' for the selected domain, including its name, type, and a 'Property View' table.

Property View	
Hub Transform Input	
Properties	
Type	Domain
Domain Name	paterva.com
WHOIS Info	
Graph Info	
Weight	0
Incoming	0
Outgoing	0

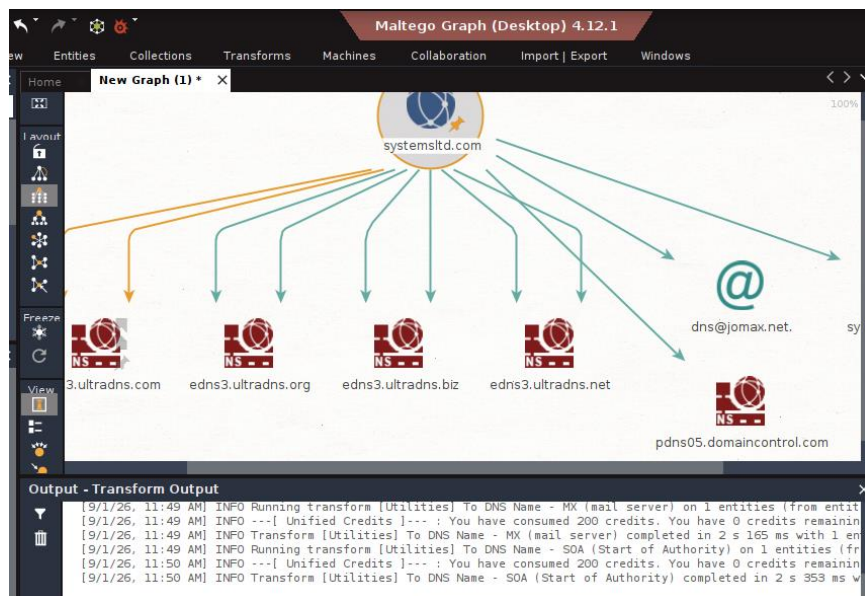
The screenshot shows the OpenCTI interface with a 'New Graph (1)' window. The graph displays a central node 'systems ltd. com' with five outgoing relationships to nodes: 'systems ltd. com. mail.protection...', 'edns3.ultradns.net', 'edns3.ultradns.biz', 'edns3.ultradns.com', and 'edns3.ultradns.org'. The relationships are color-coded: green for MX and red for NS. Below the graph, the 'Output - Transform Output' section shows a log of a 'Transform To Blacklists' operation, including a warning about an 'Unauthorized' error from the entity 'systems ltd. com'.

Executed To DNS Name - MX and To DNS Name - NS transforms on systemsltd.com. Discovered mail protection endpoint systemsltd-com.mail.protection.outlook.com and confirmed the four UltraDNS name servers.



Note on CTI Transforms: Commercial CTI (Cyber Threat Intelligence) transforms consume high API credits per query or require specialized paid subscriptions, making standard free transforms preferable for broader domain mapping.

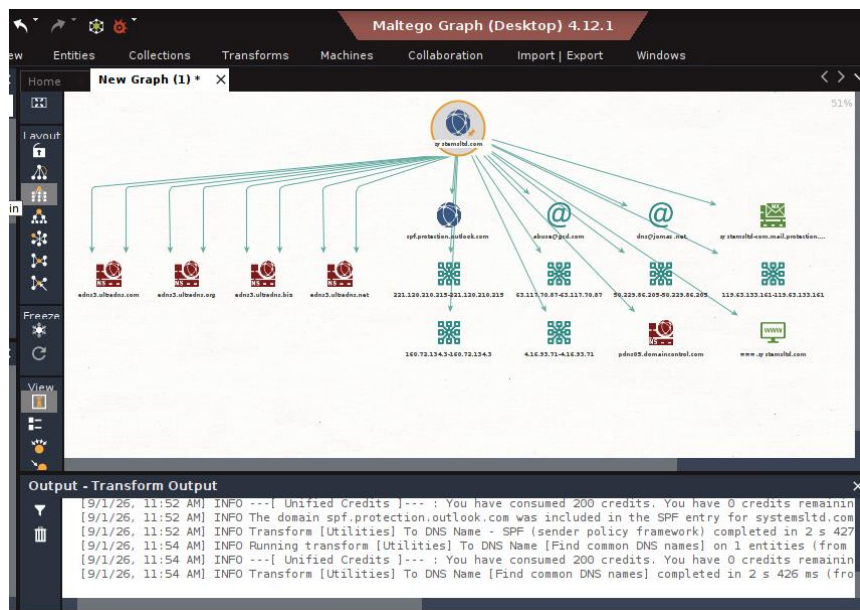
Executed To DNS Name - SOA on systemsltd.com, uncovering administrative contact emails (dns@jomax.net, abuse@gcd.com) and backup name server pdns05.domaincontrol.com.



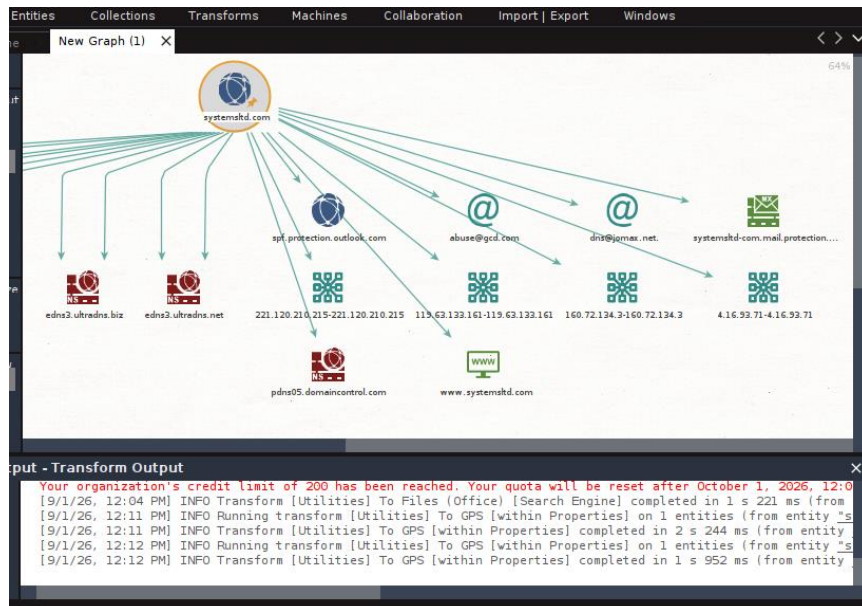
Executed To DNS Name - SPF and Quick Lookup generating netblocks (221.120.210.215-221.120.210.215, 63.117.70.87-63.117.70.87, 50.229.86.205-50.229.86.205, 119.63.133.161-119.63.133.161, 160.72.134.3-160.72.134.3, 4.16.93.71-4.16.93.71), SPF node spf.protection.outlook.com..., and web host www.systemsltd.com.



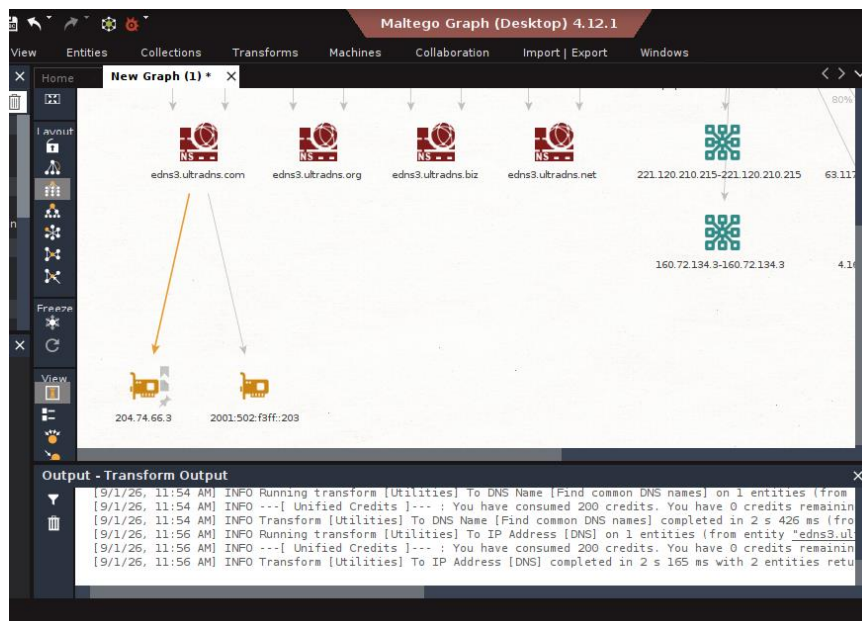
Executed To DNS Name [Find common DNS names], completing in 2.42 seconds and integrating discovered DNS endpoints into the main top-tier layout.



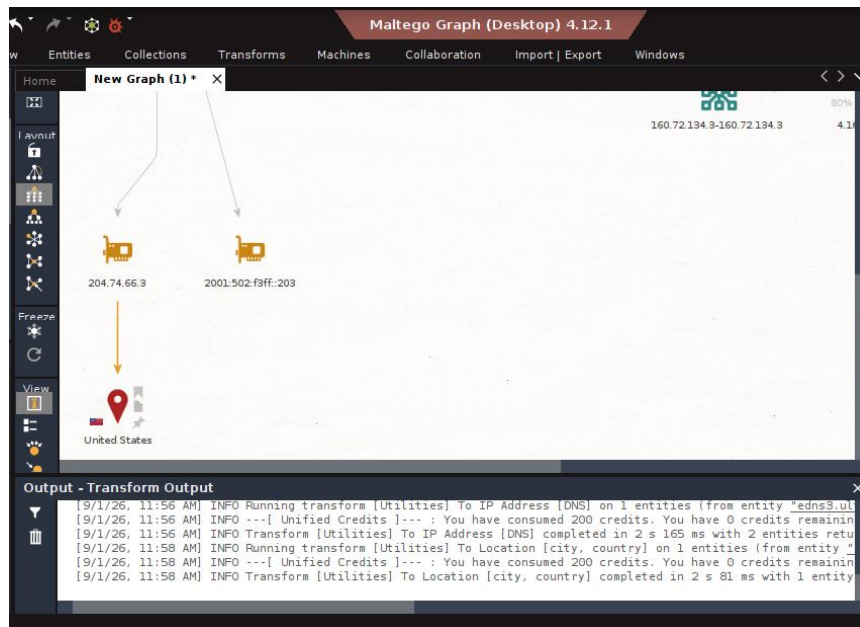
A persistent account banner noted that the organization's community credit limit of 200 had been reached, with quota reset scheduled for October 1, 2026. Despite this banner, the To GPS [within Properties] transform completed successfully, as this is a local transform that extracts existing entity properties (e.g. IP-derived coordinates) rather than querying an external paid API.



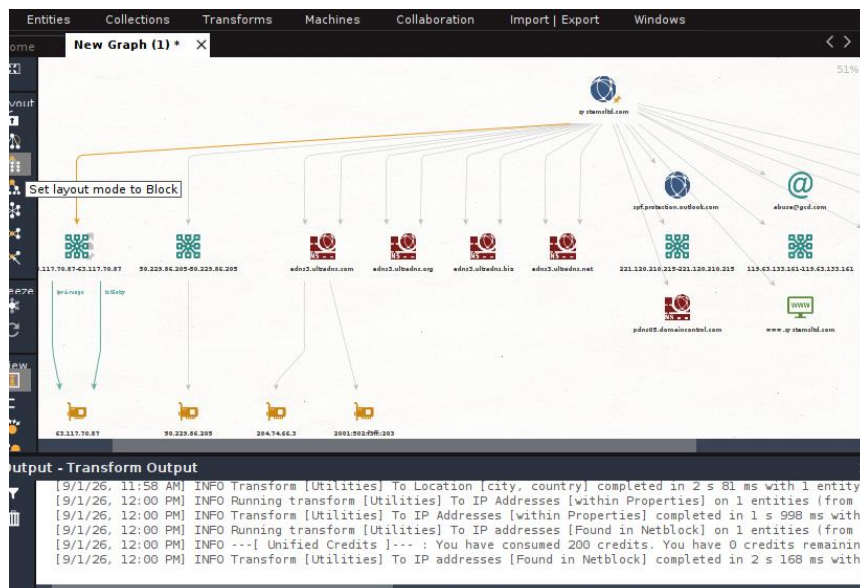
Executed To IP Address [DNS] on edns3.ultradns.com, resolving IPv4 host 204.74.66.3 and IPv6 host 2001:502:f3ff::203.



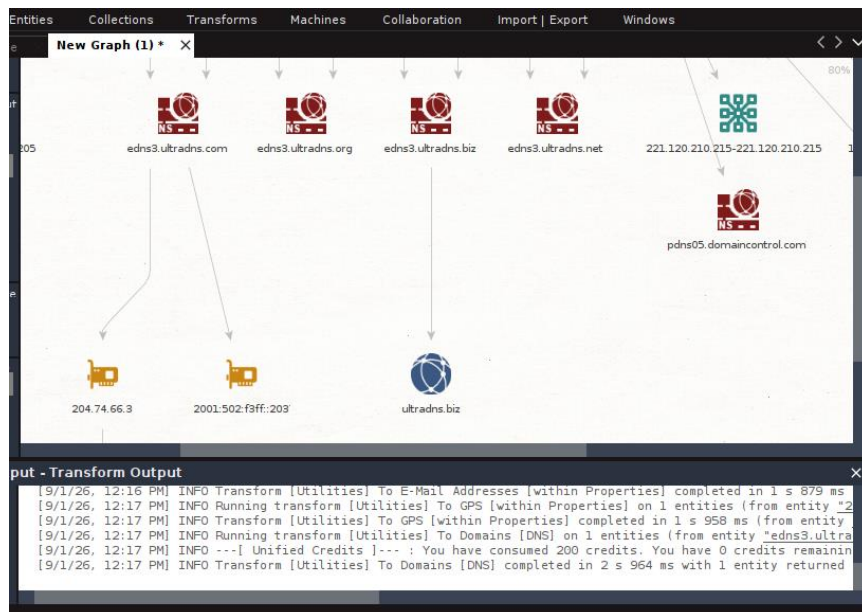
Executed To Location [city, country] on IP 204.74.66.3, mapping the physical location of the name server to the United States.



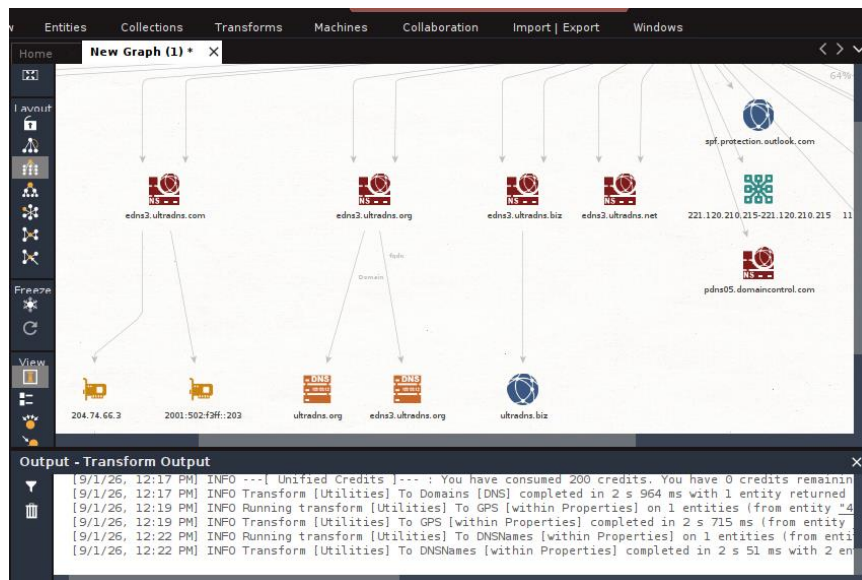
Executed To IP Addresses [Found in Netblock] on selected netblock nodes, extracting underlying host IP addresses 63.117.70.87 and 50.229.86.205.



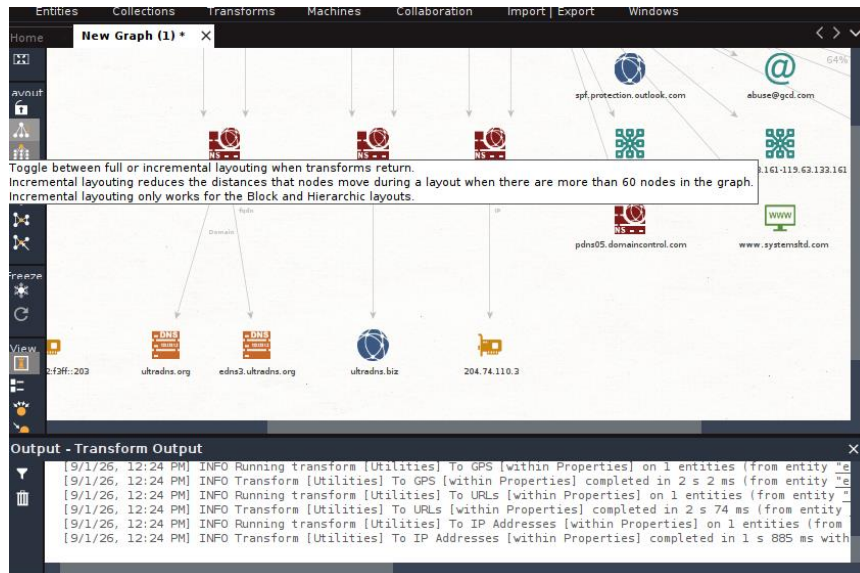
Executed To Domains [DNS] on edns3.ultradns.biz, extracting the base domain entity ultradns.biz.



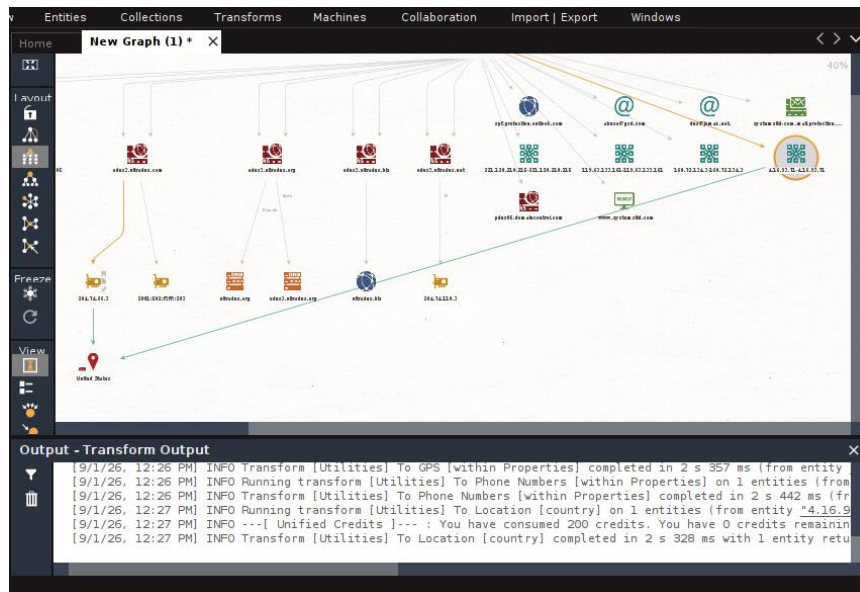
Executed To DNSNames [within Properties] on target nodes, expanding local properties into standalone DNS entities ultradns.org and edns3.ultradns.org.



Executed To IP Addresses [within Properties], extracting property-stored IP address 204.74.110.3 to the graph without API credit usage.

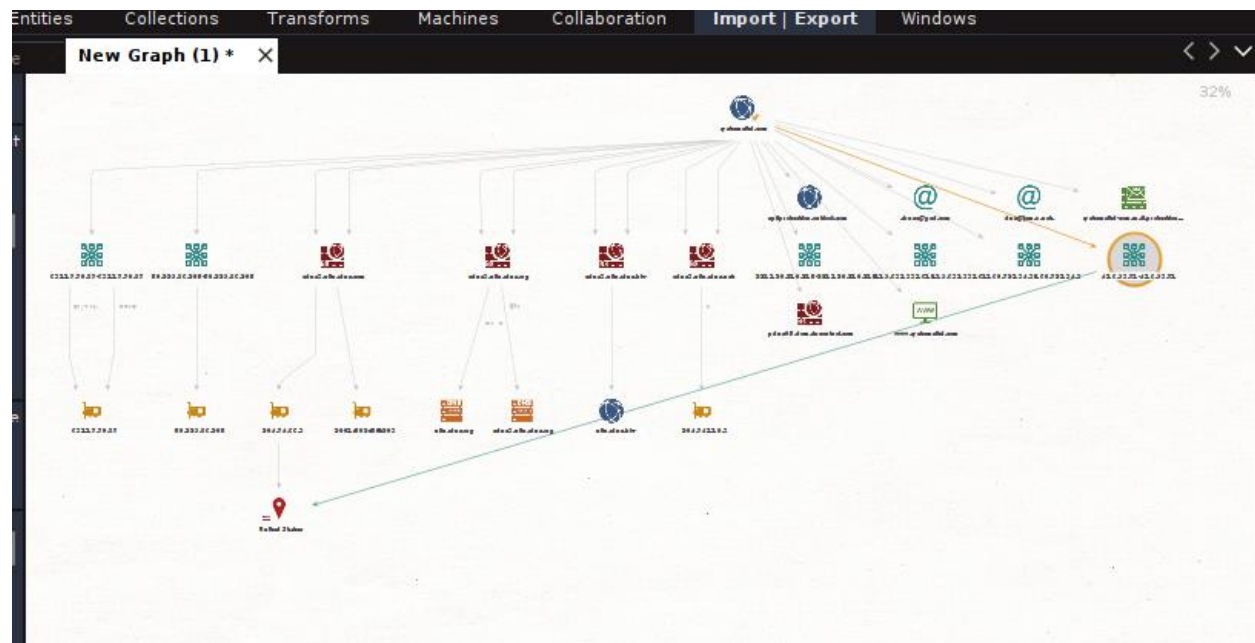


Executed To Location [country] on netblock 4.16.93.71, creating a direct relationship link to the United States location entity.



Final Topology Overview

Complete hierarchical OSINT topology graph for systemsltd.com displaying mapped mail protection servers, UltraDNS name servers, netblock ranges, IPv4/IPv6 addresses, and geographic host locations.



Conclusion

This reconnaissance exercise demonstrated the utility of Maltego CE in automating Open Source Intelligence (OSINT) collection and visualizing enterprise infrastructure topologies. Starting from a single root entity (systemsltd.com), a sequence of 18 transform operations was executed, combining commercial CTI lookups, standard DNS transforms, search-engine transforms, and local property-extraction transforms. Key findings included the organization's Microsoft Outlook-based mail protection endpoints, its four authoritative UltraDNS name servers (edns3.ultradns.com/.net/.org/.biz), SPF and SOA administrative contacts, associated netblocks and dual-stack IPv4/IPv6 host addresses, and the geographic location of hosting infrastructure in the United States. One transform (blacklist status via URLhaus) returned an authorization error, illustrating that not all commercial data sources are accessible under a free Community license — a practical constraint worth accounting for when scoping OSINT engagements. Overall, the resulting link-analysis graph provides a clear, visual blueprint of systemsltd.com's external attack surface, useful for attack-surface management and defensive security audits.